Random Forest

It is a robust, popular supervised ML algorithm used for classification and regression by creating ensemble of decision trees.
(Group)

It improves accuracy and controls overfitting by training each tree on random subsets of data (bagging) and selecting random Features For splitting nodes

Final predictions are determined by majority voting (classification) or averaging (regression)

Key Concepts and working mechanism

1) Ensemble Learning
2) Bagging (Bootstrap Aggregation)

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3) Feature Randomness

4) Voting/Averaging

Advantages

1) High accuracy
2) Prevents Overfitting
3) Robustness
4) Versatility

Disadvantages

1) <u>Reduced interpretability</u> : While a single tree is easy to interpret, a forest of hundreds of trees is harder to visualize and understand.

2) Computational cost

3) Slow prediction

Key hyperparameters

1. n_estimators - The number of trees in the Forest. Higher values increase performance but slow down computation.

2. max_features - The number of features considered for splitting a node

3. min_sample_leaf - min number of samples required to be at a leaf node.